

IMPLEMENTATION :

CSA0704 - Smart Healthcare Network Simulator

SMART HEALTHCARE NETWORK SIMULATOR

Python GUI Front-End | Protocol Analysis & Performance Assessment

TOPOLOGY

IP / VLSM

ROUTING

PING

TCP / UDP

HTTP / DNS

TRAFFIC

LINK FAILURE

RUN ALL

CISCO-STYLE COMMAND / RESULT OUTPUT

HOSPITAL CAMPUS (CORE)

Router R1 (Core)

SW1

ICU / OPD / PHARMACY

HIS SERVER (Hospital Info System)

MAN (Hospital <-> Sites)

Router R2 / Router R3

SW2 / SW3

Remote Clinic (Telemedicine) / Diagnostic Lab / Radiology

Video consult PCs / Imaging workstations

Patient monitoring IoT / IoT diagnostic sensors

Design : Hybrid star topology (redundant core, segmented departments)

Devices : 3 Routers, 3 Switches, ICU/OPD/Pharmacy endpoints, 1 HIS Server, Remote Clinic + Diagnostic Lab endpoints, IoT medical sensors (vitals monitors, imaging devices)

Rationale : ICU and OPD are isolated into separate broadcast domains for security (HIPAA-style data isolation) and to protect latency-sensitive ICU/IoT traffic from bulk OPD traffic. WAN links connect the Remote Clinic and Diagnostic Lab for telemedicine and centralized diagnostics.

TOPOLOGY COMPLETED

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IP ADDRESSING PLAN (VLSM)

Base network block : 192.168.20.0/24

Segment	Hosts Needed	Subnet	Usable Range
OPD (Out-Patient Dept.)	60	192.168.20.0/26	.1 - .62
Pharmacy	30	192.168.20.64/27	.65 - .94
ICU (critical, low-latency)	14	192.168.20.96/28	.97 - .110
HIS Server subnet	14	192.168.20.112/28	.113 - .126
Remote Clinic (Telemedicine)	14	192.168.20.128/28	.129 - .142
Diagnostic Lab / Radiology	14	192.168.20.144/28	.145 - .158
R1 <-> R2 WAN link	2	192.168.20.160/30	.161 - .162
R1 <-> R3 WAN link	2	192.168.20.164/30	.165 - .166

VLSM CHECK : No overlapping subnets, all ranges within /24 : PASS

ADDRESS UTILIZATION: 218 usable hosts allocated of 254 available : PASS

RESULT: IP / VLSM PLAN VERIFIED

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CISCO-STYLE COMMAND / RESULT OUTPUT

ROUTING TABLE VERIFICATION (OSPF Area 0)

```
Router R1 (Core)
O 192.168.28.128/28 [110/20] via 192.168.28.162, R2
O 192.168.28.144/28 [110/20] via 192.168.28.166, R3
C 192.168.28.0/26 directly connected, SM1 (OPD)
C 192.168.28.96/28 directly connected, SM1 (ICU)
C 192.168.28.112/28 directly connected, SM1 (HIS Server)

Router R2 (Remote Clinic)
O 192.168.28.0/24 summary [110/20] via 192.168.28.161, R1

Router R3 (Diagnostic Lab)
O 192.168.28.0/24 summary [110/20] via 192.168.28.165, R1
```

OSPF NEIGHBOR STATUS

```
R1 <-> R2 : FULL/DR
R1 <-> R3 : FULL/DR
```

CONVERGENCE TIME : 3.2 sec

ROUTE VERIFICATION : ALL ROUTES REACHABLE : PASS

RESULT: ROUTING VERIFIED

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Router R3 (Diagnostic Lab)
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```

OSPF NEIGHBOR STATUS

```
R1 <-> R2 : FULL/DR
R1 <-> R3 : FULL/DR
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CONVERGENCE TIME : 3.2 sec

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CISCO-STYLE COMMAND / RESULT OUTPUT

PING CONNECTIVITY TEST

```
ICU-PC (192.168.20.98) -> HIS Server (192.168.20.113) : SUCCESS (avg 2 ms)
OPD-PC (192.168.20.5) -> HIS Server (192.168.20.113) : SUCCESS (avg 4 ms)
Pharmacy-PC (192.168.20.66) -> HIS Server (192.168.20.113) : SUCCESS (avg 3 ms)
HIS Server -> Remote Clinic (192.168.20.130) : SUCCESS (avg 21 ms)
HIS Server -> Diagnostic Lab (192.168.20.146) : SUCCESS (avg 19 ms)
Remote Clinic IoT Monitor -> HIS Server : SUCCESS (avg 24 ms)
```

```
PACKET LOSS : 0%
END-TO-END REACHABILITY: ALL NODES REACHABLE : PASS
```

RESULT: PING CONNECTIVITY VERIFIED

PING COMPLETED

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CISCO-STYLE COMMAND / RESULT OUTPUT

PROTOCOL TEST (TCP vs UDP) - used for Electronic Health Records vs Live Vitals

TCP (used for HIS / EHR record transfer)

```
Connection : Established (3-way handshake)
Delivery : Reliable, ordered, retransmits lost segments
Use case : Patient record sync, prescription updates, billing
Status : PASS
```

UDP (used for real-time patient vitals / IoT monitoring streams)

```
Connection : Connectionless
Overhead : Low (no handshake, no retransmission)
Use case : ICU heart-rate / SpO2 streaming, video consult (telemedicine)
Status : PASS
```

```
TCP -> reliable, ordered delivery, best for critical medical records
UDP -> low-latency, tolerates minor loss, best for live vitals/streaming
```

RESULT: TCP / UDP ANALYSIS COMPLETED

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HIS SERVER

IP Address : 192.168.20.113

HTTP SERVICE (Hospital Patient Portal)

Status : ON

Request : http://his.hospital.local

Result : Web response received (Patient Portal loaded)

DNS SERVICE

Status : ON

his.hospital.local -> 192.168.20.113

clinic.hospital.local -> 192.168.20.130

lab.hospital.local -> 192.168.20.146

Result : Name resolution successful for all hospital domains

RESULT: HTTP / DNS VERIFIED

HTTP / DNS COMPLETED

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PERFORMANCE / TRAFFIC TEST

Condition	Latency	Throughput	Packet Loss
Normal	16 ms	95 Mbps	1%
Peak OPD Hours	33 ms	80 Mbps	4%
Emergency Surge	62 ms	58 Mbps	10%

Observation:
As OPD/ICU traffic increases (peak hours, emergency surge), latency and packet loss rise while available throughput drops. This is critical for ICU vitals monitoring and telemedicine, where sustained high latency or packet loss can delay alerts to clinical staff.

Recommendation:
QoS should prioritize ICU/IoT vitals traffic (UDP, low latency) above routine OPD/administrative traffic (TCP) during peak/congested periods.

RESULT: PERFORMANCE ANALYSIS COMPLETED

TRAFFIC COMPLETED

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CISCO-STYLE COMMAND / RESULT OUTPUT

LINK FAILURE TEST (Critical Path: ICU Switch <-> Core Router)

BEFORE FAILURE

ICU-PC -> HIS Server : SUCCESS

LINK DOWN

R1 G0/1 ----- X ----- SW1 (ICU)

DURING FAILURE

ICU-PC -> HIS Server : REQUEST TIMED OUT

ALERT : ICU vitals monitoring stream INTERRUPTED

RECOVERY

Backup path activated via redundant OSPF route

Link restored

ICU-PC -> HIS Server : SUCCESS

IMPACT NOTE:

In a real hospital deployment, this link should have hardware redundancy (dual NICs / secondary switch uplink) since ICU connectivity loss directly affects patient safety monitoring.

RESULT: FAILURE AND RECOVERY VERIFIED

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CISCO-STYLE COMMAND / RESULT OUTPUT

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SMART HEALTHCARE NETWORK - FINAL VERIFICATION

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IPv4 / VLSM : PASS

Routing (OSPF) : PASS

Ping Connectivity : PASS

TCP : PASS

UDP : PASS

HTTP : PASS

DNS : PASS

Traffic Analysis : TESTED

Link Failure / Recovery : TESTED

Normal : 16 ms | 95 Mbps | 1% loss

Peak OPD Hours : 33 ms | 88 Mbps | 4% loss

Emergency Surge : 62 ms | 58 Mbps | 10% loss

FINAL STATUS: SMART HEALTHCARE NETWORK TESTING COMPLETED

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ALL TESTS COMPLETED